

HR Analytics - Employee Attrition Prediction Report

Objective

Analyze employee attrition, identify key factors influencing resignation, build a predictive classification model, explain predictions using SHAP, and create an interactive Power BI dashboard.

Dataset

IBM HR Analytics Employee Attrition dataset containing employee demographic, job, compensation, and performance information.

Tools Used

Google Colab (Python), Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, SHAP, Power BI Desktop.

Methodology

1. Imported and explored the dataset. 2. Performed EDA using charts for department, salary, overtime, promotions, age, and job role. 3. Encoded categorical variables. 4. Split the data into training (80%) and testing (20%). 5. Trained a Logistic Regression or Decision Tree classifier. 6. Evaluated the model using accuracy, confusion matrix, precision, recall, and F1-score. 7. Used SHAP values to explain feature importance. 8. Built an interactive Power BI dashboard.

Exploratory Analysis

Visualizations included department-wise attrition, job role vs attrition, gender vs attrition, overtime vs attrition, monthly income vs attrition, age distribution, promotion analysis, and a correlation heatmap.

Machine Learning

Target variable: Attrition. Features: Remaining employee attributes. Evaluation metrics: Accuracy, confusion matrix, precision, recall, and F1-score.

SHAP Analysis

SHAP was used to identify the features contributing most to attrition predictions and improve model interpretability.

Power BI Dashboard

Dashboard includes KPI cards (Total Employees, Employees Left, Average Age, Average Income), charts for attrition factors, and slicers for Department, Gender, Job Role, and Marital Status.

Recommendations

Improve promotion opportunities; review compensation for lower salary bands; reduce excessive overtime; improve work-life balance; provide career development; recognize employee performance; conduct regular engagement surveys; implement targeted retention strategies.

Conclusion

The project demonstrates how data analytics and machine learning can identify attrition drivers, predict employee resignation, and support HR decision-making.